Task 3 Report: Public Email Intelligence Collector

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Authorized scope: A synthetic lab website served from localhost (127.0.0.1). All emails are fake (site.test).

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1. Executive Summary

I built a crawler that finds email addresses on web pages. It starts at one URL, follows links up to a chosen depth, and only visits pages on the same site. It reads emails from page text, mailto: links, and hidden forms like name [at] site [dot] test. Each email is saved once, with every page it appeared on, a timestamp, and a confidence level.

I tested it on a small synthetic website. At depth 2 it found **7 unique emails on 5 pages**. At depth 3 it found **8 emails on 6 pages**. It never visited the page blocked by robots.txt and never requested the out-of-scope link. All **26 automated tests pass**, and the dependency check found no known vulnerabilities.

Main conclusion: the tool meets all 10 Task 3 requirements and stays inside its authorized scope.

2. Scope and Rules Followed

- Only a local synthetic site was crawled, with fake emails.
- Non-local targets are refused unless the user adds --authorized.
- robots.txt is respected.
- There is a minimum delay of 0.5 seconds between requests.
- Only same-host pages are visited. Out-of-scope links and redirects are never requested.
- Only http and https URLs are accepted. URLs with embedded credentials are rejected.
- No real person's email address was collected.

3. How the Tool Works

Module	What it does
validators.py	Checks the start URL and the scope. Rejects lookalike domains
crawler.py	Fetches pages, parses HTML, extracts emails, follows in-scope links
main.py	Command line, safety checks, terminal table, JSON output

Run command: python3 -m src.main --url http://127.0.0.1:8000/ --depth 2 --delay 0.5

Every result uses the same record structure: type, value, source, timestamp, confidence, scope, metadata.

4. Requirements Coverage

#	Requirement	Status	Evidence
1	URL crawling	Done	5 pages at depth 2, 6 pages at depth 3
2	Configurable crawl depth	Done	--depth option. Depth 3 found deep@site.test, depth 2 did not
3	HTML parsing	Done	BeautifulSoup, including a deliberately broken page (contact.html)
4	Email-pattern identification	Done	Text, mailto and obfuscated emails
5	Duplicate removal	Done	info@site.test is on 2 pages but appears once
6	Source URL recording	Done	metadata.source_urls

7	Timestamp recording	Done	timestamp field on every record
8	Rate limiting	Done	Minimum 0.5 second delay, enforced by the CLI
9	Scope enforcement	Done	Out-of-scope link and redirect never requested
10	Structured output	Done	output/127.0.0.1_depth2.json and depth3.json

5. Findings

5.1 Test website

The lab site has 8 files. It includes a robots.txt that blocks /private/, a malformed page, an obfuscated email, an image filename that looks like an email, and a link to an outside site.



5.2 Depth 2 result

7 unique emails, 5 pages crawled, 1 out-of-scope link skipped, 1 page skipped by robots.txt.



5.3 Depth 3 result

8 unique emails, 6 pages crawled. The extra email is deep@site.test, which is 3 clicks from the start page.



5.4 Detection details

Case	What happened	Case

hr [at] site [dot] test	Found as hr@site.test, confidence **medium**, method obfuscated	hr [at] site [dot] test
mailto: link	sales@site.test found, method mailto	mailto: link
Broken HTML	support@site.test found on contact.html without a crash	Broken HTML
logo@2x.png	Looks like an email but is an image filename. Filtered out	logo@2x.png
/private/secret.html	Blocked by robots.txt. Never fetched, so secret@site.test is not in the results	/private/secret.html
info@site.test	On two pages, saved once with both source URLs	info@site.test



5.5 Safety checks

Three unsafe commands were refused before any request was sent:

- A non-local URL without --authorized
- An ftp:// URL
- A delay of 0.1 seconds (below the 0.5 second minimum)

6. Verified Facts vs Inference

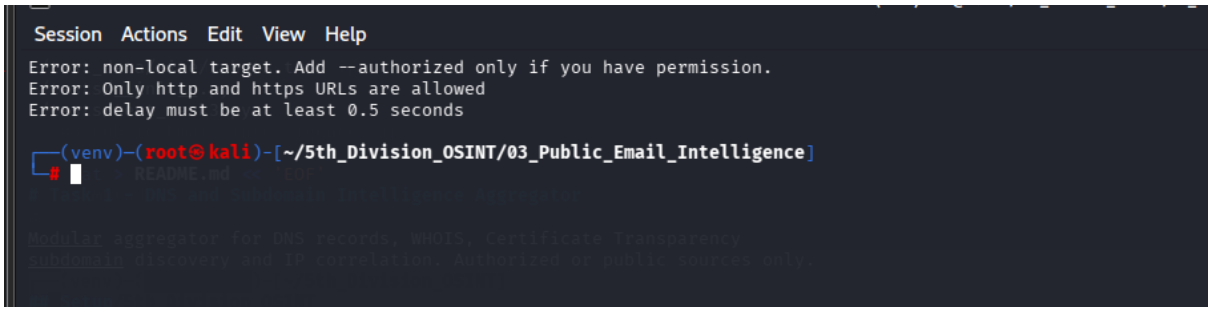
Type	Meaning
Verified fact	An email in page text or a mailto: link, with the page URL as proof
Lower confidence	An obfuscated email rebuilt from [at] and [dot]. It is a reconstruction
Not proven	That an address is active, or belongs to a specific person

7. Testing

There are 26 offline tests. The network is mocked, so they are repeatable and send no real requests. They cover:

- Valid and invalid URLs
- Scope checks and lookalike domains
- Text, mailto and obfuscated emails
- Image-name filtering
- Malformed and empty HTML
- Relative link resolution
- Source recording and duplicate merging
- Depth zero
- Out-of-scope links and redirects
- Timeouts and HTTP errors
- Non-HTML pages
- Page limit
- robots.txt
- Record format

Run: python3 -m pytest tests -v
Results saved in: output/test_results.txt



8. Security Controls

- URL validation (http and https only, no embedded credentials, length limit)
- Strict scope check with lookalike protection
- robots.txt respected
- Rate limit, timeouts, page limit and depth limit

- Explicit --authorized flag for non-local targets
- No credentials or API keys in the code
- One failing page never stops the crawl
- Dependencies checked with pip-audit. A first run flagged only pip 26.1.1 (the installer, not project code). After upgrading to 26.2.1, the result was: no known vulnerabilities found



9. Limitations and Known Issues

- Tested only on a local synthetic site, not on large real websites.
- Pages built with JavaScript are not rendered, so emails added by scripts are missed.
- Obfuscation detection covers only [at]/(at) and [dot]/(dot).
- Redirects are followed only inside scope, one hop at a time.
- Found emails are personal data. Only public, in-scope pages should be crawled.

10. False-Positive Considerations

- Strings like logo@2x.png look like emails. A file-extension filter removes them.
- Role addresses (info@, support@) may be shared mailboxes, not individuals.
- Obfuscated results are less certain, so they are marked medium.

11. Conclusion

The tool meets all 10 Task 3 requirements. It respects scope, robots.txt and rate limits, removes duplicates, records the source and time of every email, and was tested only on authorized synthetic data. The main limits are that it does not run JavaScript and was tested on a small site.

12. Evidence Index

File	Purpose
output/127.0.0.1_depth2.json	Depth 2 results
output/127.0.0.1_depth3.json	Depth 3 results
output/test_results.txt	26 passing tests
output/pip_audit.txt	Dependency check
test_data/site/	Synthetic test website
screenshots/	Evidence images 1 to 7